

MULTI-ANATOMY ADAPTATION AND CONFIGURATION EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device is configured to operate across multiple anatomical environments while maintaining a common sensing and computational architecture. The device may be adapted for use within vaginal, anal, or other intraluminal environments without requiring fundamental redesign of the core distance sensing, compliance sensing, or profile generation systems.

Adaptation to different anatomical environments may be achieved through variation of insertable length, diameter, taper, flexibility, surface compliance, and distal tip geometry. Devices may be produced in multiple physical sizes, multiple stiffness tiers, or adjustable configurations while preserving standardized sensing operation and data interpretation frameworks.

In certain embodiments, the device comprises modular or interchangeable components. For example, interchangeable distal tips, sleeves, expandable regions, or insertable bodies may be coupled to a common handle and electronics platform. Modular adaptation may enable a single controller architecture to support multiple anatomical configurations through replaceable mechanical assemblies.

The device may further adapt operational behavior based on selected mode or detected use context. Software configuration profiles may adjust inflation limits, safety thresholds, sensing thresholds, stabilization requirements, and guidance workflows according to the intended measurement environment. In certain embodiments, inflation rate, maximum allowable pressure, and allowable expansion range are mode-dependent to maintain safe operating conditions.

Distance sensing algorithms may compensate for differences in curvature, reflectivity, moisture, acoustic coupling, and tissue response encountered across anatomical environments. In various embodiments, the computational system applies filtering, multi-sample averaging, and confidence scoring to improve robustness across variable measurement conditions.

Compliance measurement may likewise be adapted by selecting different sensing regions, different inflatable geometries, or different pressure sensing distributions. Some embodiments may employ localized cuffs for targeted measurement, while other embodiments may employ extended sensing coverage for broader characterization. In certain embodiments, compliance outputs are normalized into standardized indices to remain comparable across configurations and sensing variants.

In anatomy-agnostic implementations, sensing data is interpreted without reliance on anatomical labeling. In such embodiments, the computational system derives structural and compliance characteristics from measurement data directly and stores the results as abstracted parameters, thereby preserving flexibility for future luminal applications.

Multi-anatomy configurations may also include accessory and ergonomic variations such as alternative handle orientations, insertion angle aids, reference flanges of differing geometries, or positioning guidance features intended to improve usability and repeatability across contexts.

Accordingly, the disclosed architecture provides a unified intraluminal sensing platform capable of serving multiple anatomical applications while maintaining common calibration standards, shared computational infrastructure, consistent safety constraints, and comparable user profiling methodologies.